

OPTRONIC SENSORS: Applications In Detonics



Dr Anil Kumar Maini was formerly outstanding scientist and director at Laser Science and Technology Centre (DRDO)



Nakul Maini is post graduate in optical engineering from University of Bristol (UK), currently working as senior engineer (product development) at Tek Xplore, New Delhi

Optronic sensors are at the core of a large number of optronic instrumentation setups used for a variety of applications in detonics. Some representative examples include interferometric sensors for blast pressure measurements, visar (an acronym for velocity interferometer system for any reflector) sensors for measurement of surface velocity of solids, spectrometric and spectroscopic sensors for measurement of fireball temperature, chirped-fibre Bragg grating (CFBG) sensors for measurement of detonation velocity, and triangulation and time-of-flight sensors for measurement of position and displacement caused by blast loading and laser velocity screen for measurement of velocity and aiming accuracy of projectiles of different calibres.

Beginning with a brief introduction to various techniques and related sensors, this concluding part of the article focuses on optronic triangulation and time-of-flight sensors for position and displacement measurement and velocity screen used for measurement of projectile velocity.

Common optronic sensors with applications in detonics depend for their operation on one or more of the following techniques. These include laser interferometry, laser velocimetry, laser spectroscopy, spectrometry, laser triangulation and laser time-of-flight ranging. Laser velocity screen is also based on time-of-flight principle, where time taken by the projectile to travel a known distance is used to compute projectile velocity.

Interferometric sensors

Interferometric sensors have demonstrated tremendous potential in the measurement of blast parameters such as peak-over pressure, pressure-time history, detonation velocity, shock-wave phenomenon and so on. Fabry-Perot cavity sensors, for example, can be used for the measurement of peak-over pressure and pressure-time history resulting from explosive blasts.

A pressure sensor comprises a fibre-optic probe (Fig. 1) and an interrogation system for Fabry-Perot cavity-based pressure sensor. The system involves exciting the fibre-optic probe

with a broadband laser radiation through a fibre-optic coupler. A part of laser radiation enters Fabry-Perot interferometer terminated in a pressure-sensitive diaphragm. Two laser beams—one reflecting from the front face and the other reflecting from the diaphragm interfere—produce a fringe pattern. Interference pattern is processed further to extract pressure information in terms of peak-over pressure and pressure-time history.

Visar uses laser interferometry to measure surface velocity of solids moving at high velocities. It is a useful diagnostic tool for measuring shock-wave velocity of matter in extreme conditions, and is used in many laboratories as a tool to study high-pressure shock. Visar measures changes in Doppler shift of light reflected off a moving surface to compute velocity and velocity-time history.

Doppler shift appears in the form of fringes. Measurements made at different times give velocity-time history, which, in turn, gives the shock-wave profile. Velocity is computed as the product of magnitude of fringe shift and a parameter called velocity per fringe.

Chirped-fibre Bragg grating sensors

A CFBG is a grating sensor having a periodic pitch. Grating is characterised by three important parameters: grating period, refrac-

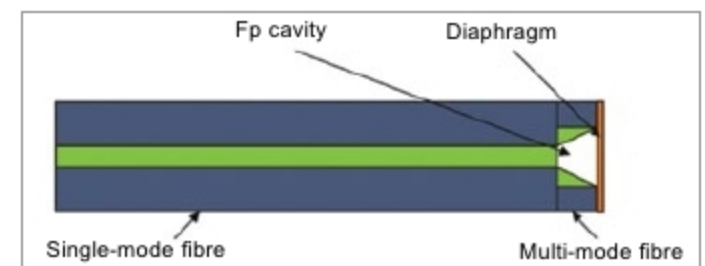


Fig. 1: Pressure sensor

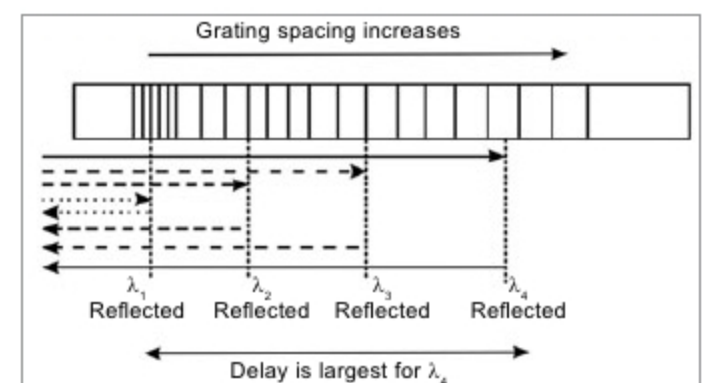


Fig. 2: Schematic of grating in which grating period is increasing linearly with length of grating

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